Experiment No : 4

Title: Implement Database Management using arrays of structures with operations Create, Display, Modify,Search and Sort.

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Program:

```
#include <stdio.h>

struct Student
{
   int id;
   char name[20];
   int marks;
};

int main()
{
   struct Student s[50], temp;
   int n = 0, ch, i, j, id, found;

   do
   {
      printf("\n1.Create\n2.Display\n3.Modify\n4.Search\n5.Sort\n6.Exit");
      printf("\nEnter choice: ");
      scanf("%d", &ch);

      switch(ch)
      {
         case 1:
            printf("Enter number of students: ");
```

```
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
       printf("Enter ID, Name and Marks: ");
       scanf("%d %19s %d",
            &s[i].id, s[i].name, &s[i].marks);
    }
    break;

case 2:
    for(i = 0; i < n; i++)
    {
       printf("\n%d %s %d",
             s[i].id, s[i].name, s[i].marks);
    }
    break;

case 3:
    printf("Enter ID to modify: ");
    scanf("%d", &id);
    found = 0;

    for(i = 0; i < n; i++)
    {
       if(s[i].id == id)
       {
          printf("Enter new marks: ");
          scanf("%d", &s[i].marks);
          found = 1;
          break;
       }
```

```
    }

    if(found == 0)
       printf("Record not found");
    break;

case 4:
    printf("Enter ID to search: ");
    scanf("%d", &id);
    found = 0;

    for(i = 0; i < n; i++)
    {
       if(s[i].id == id)
       {
          printf("%d %s %d",
                s[i].id, s[i].name, s[i].marks);
          found = 1;
          break;
       }
    }

    if(found == 0)
       printf("Record not found");
    break;

case 5:
    for(i = 0; i < n - 1; i++)
    {
       for(j = 0; j < n - i - 1; j++)
       {
          if(s[j].id > s[j+1].id)
```

```
                {
                    temp = s[j];
                    s[j] = s[j+1];
                    s[j+1] = temp;
                }
            }
        }

        printf("Records sorted by ID");
        break;

    case 6:
        printf("Exiting...");
        break;

    default:
        printf("Invalid choice");
  }

} while(ch != 6);

return 0;
}
```

Code :-

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 1
Enter number of students: 4
Enter ID, Name and Marks: 1 medhavi 99
2 ashwika 99
3 shamiksha 99
4 ishita 99
Enter ID, Name and Marks: Enter ID, Name and Marks: Enter ID, Name and
Marks:
```

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 2

1 medhavi 99
2 ashwika 99
3 shamiksha 99
4 ishita 99
```

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 3
Enter ID to modify: 2
Enter new marks: 100
```

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 4
Enter ID to search: 4
4 ishita 99
```

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 5
Records sorted by ID
```

```
1.Create
2.Display
3.Modify
4.Search
5.Sort
6.Exit
Enter choice: 6
Exiting...
```